



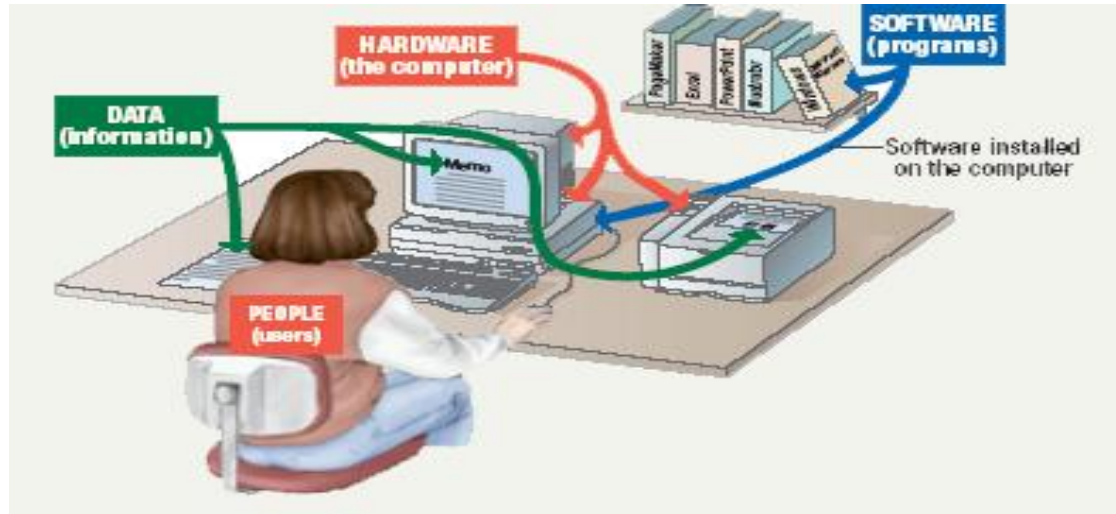
Looking Inside the Computer



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Parts of the Computer System

- ▶ Computer systems have four parts
 - ▶ Hardware
 - ▶ Software
 - ▶ Data
 - ▶ User



Parts of the Computer System

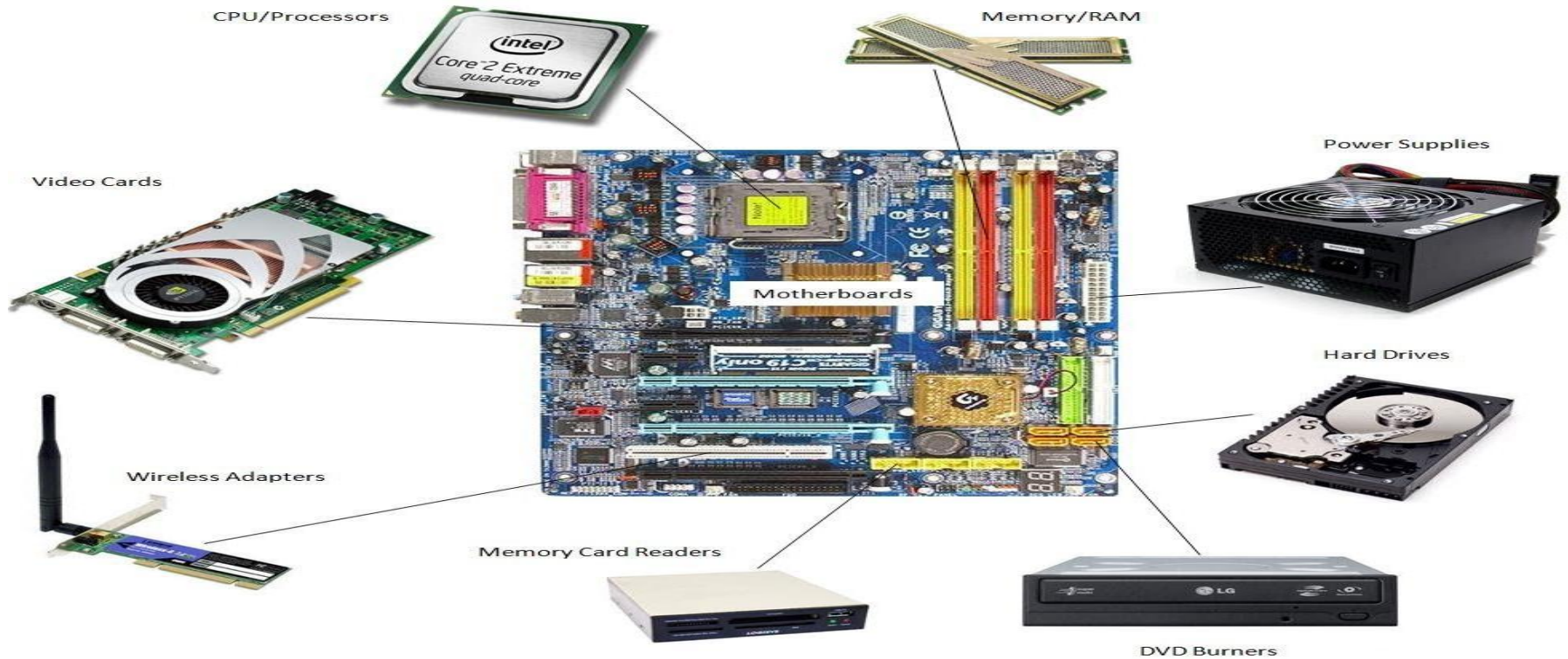
- ▶ **Hardware**

- ▶ Mechanical devices in the computer
- ▶ Anything that can be touched

- ▶ **Software**

- ▶ Tell the computer what to do
- ▶ Also called a program
- ▶ Thousands of programs exist

Hardware



Parts of the Computer System

- ▶ **Data**

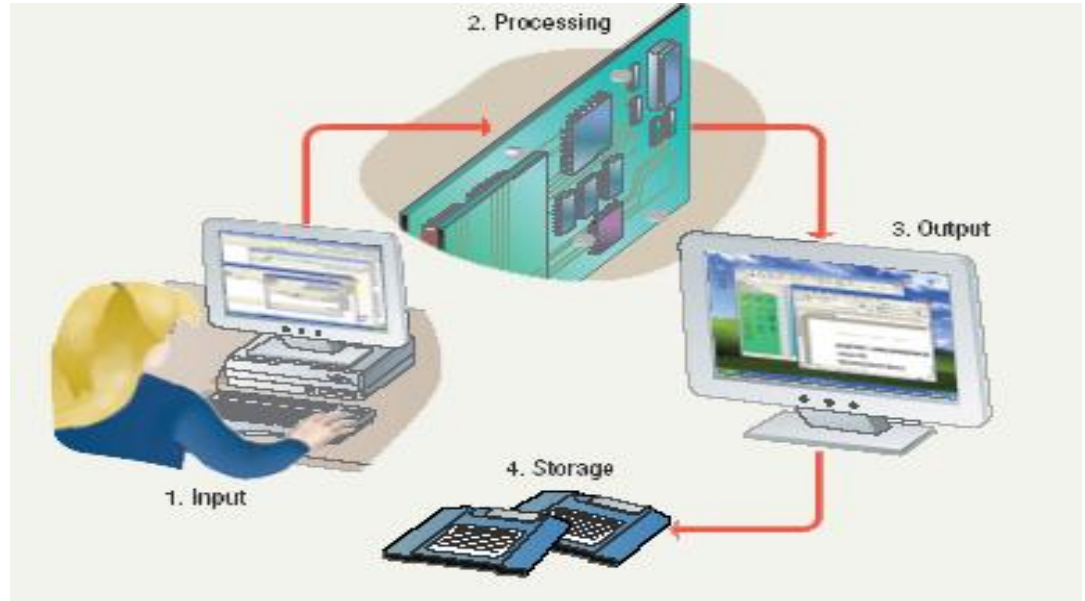
- ▶ Pieces of information
- ▶ Computer organize and present data

- ▶ **Users**

- ▶ People operating the computer
- ▶ Most important part
- ▶ Tell the computer what to do

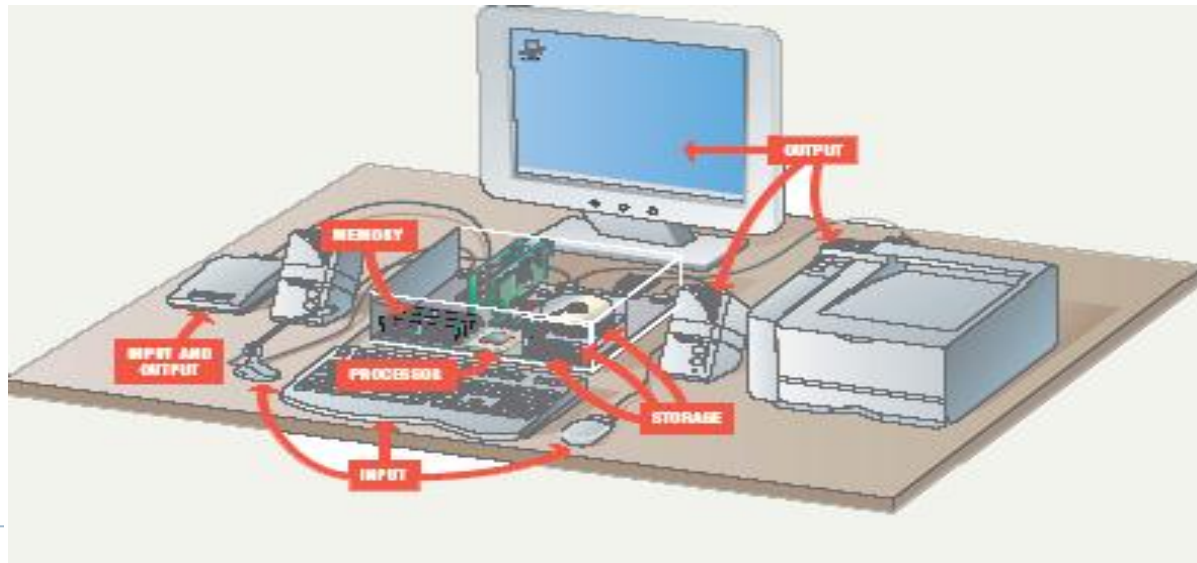
Information Processing Cycle

- ▶ Steps followed to process data
- ▶ Input
- ▶ Processing
- ▶ Output
- ▶ Storage



Essential Computer Hardware

- ▶ Computers use the same basic hardware
- ▶ Hardware categorized into four types



Essential Computer Hardware

- ▶ Input and output devices
 - ▶ Allows the user to interact
 - ▶ Input devices accept data
 - ▶ Keyboard, mouse
 - ▶ Output devices deliver data
 - ▶ Monitor, printer, speaker
 - ▶ Some devices are input and output
 - ▶ Touch screens

Input Devices

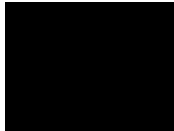
- ▶ Enable **human-computer interaction**
- ▶ Voice Input
 - ▶ Smart assistants (e.g., Alexa, Google Assistant, Siri)
- ▶ Natural Language Input
 - ▶ Chat interfaces powered by AI (like ChatGPT or Copilot)
 - ▶ AI writing tools that understand and generate text
- ▶ Context-Aware Input
 - ▶ Location, user behavior, and time influencing AI



Prompt Engineering: A Modern Input Method

- ▶ The strategic crafting of input text or instructions to guide an AI system's response with precision and relevance.
- ▶ Treats **textual prompts** as dynamic input to language models.
- ▶ Bridges human **intent** and machine **interpretation**.
- ▶ Examples of Input Prompts
 - ▶ “Summarize the role of AI in education system.”

How does AI smell?



https://www.youtube.com/watch?v=VFbYadm_mrw



<https://www.youtube.com/shorts/nXN7l6GjLH8>

Today's Fictions Tomorrow's Innovations

Output Devices: Sharing Results from Processing

- ▶ Devices that present processed data from the computer to the user in readable or tangible form.
- ▶ Support visualization of results from **data analyses** to **AI responses**
- ▶ Examples
 - ▶ **Neuralink**: Allows users to move cursors or type using thought alone
 - ▶ **BCI Wheelchairs**: Navigate based on brain signals

What if we attempt to produce a
human organ as an output of a
technological?

<https://www.youtube.com/watch?v=bX3C201O4MA>

Essential Computer Hardware

- ▶ **Processing devices**
 - ▶ Brains of the computer
 - ▶ Carries out instructions from the program
 - ▶ Manipulate the data
 - ▶ Most computers have several processors
 - ▶ Central Processing Unit (CPU)
 - ▶ Secondary processors
 - ▶ Processors made of silicon and copper

Essential Computer Hardware

- ▶ **Memory devices**

- ▶ Stores data or programs
- ▶ Random Access Memory (RAM)
 - ▶ Volatile
 - ▶ Stores current data and programs
 - ▶ More RAM results in a faster system
- ▶ Read Only Memory (ROM)
 - ▶ Permanent storage of programs
 - ▶ Holds the computer boot directions

Essential Computer Hardware

- ▶ Storage devices
 - ▶ Hold data and programs permanently
 - ▶ Different from RAM
 - ▶ Magnetic storage
 - ▶ Floppy and hard drive
 - ▶ Uses a magnet to access data
 - ▶ Optical storage
 - ▶ CD and DVD drives
 - ▶ Uses a laser to access data

Can we store data outside computing devices?

<https://www.youtube.com/watch?v=r8qWc9X4f6k>

Software Runs The Machine

- ▶ Tells the computer what to do
- ▶ Reason people purchase computers
- ▶ Two types
 - ▶ System software
 - ▶ Application software

Software Runs The Machine

- ▶ System software
 - ▶ Most important software
 - ▶ Operating system
 - ▶ Windows XP
 - ▶ Network operating system (OS)
 - ▶ Windows Server 2003
 - ▶ Utility
 - ▶ Symantec AntiVirus



Software Runs The Machine

- ▶ Application software

- ▶ Accomplishes a specific task
- ▶ Most common type of software
 - ▶ MS Word, Mozilla Firefox etc.
- ▶ Covers most common uses of computers
- ▶ Various Games
 - ▶ Candy Crash, Racing Game, Temple run games etc.
- ▶ Communication Software
 - ▶ Messenger, Skype, viber etc.

Computer users

- ▶ Role depends on ability
 - ▶ Setup the system
 - ▶ Install software
 - ▶ Manage files
 - ▶ Maintain the system
- ▶ “Userless” computers
 - ▶ Run with no user input
 - ▶ Automated systems



End of the Lecture